MIL–D–24/9B
10 APRIL 1964
SUPERSEDING
MIL–D–24/9A
1 JUNE 1961

MILITARY SPECIFICATION

DYNAMOTOR, TYPE DY–134/GRC–9X

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers the detail requirements for type DY–134/GRC–9X dynamotors.

1.2 The dynamotor covered herein is not intended for use over 10,000 feet.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

MIL–D–24 —Dynamotors, General Specification for.
MIL–I–11748—Interference Reduction for Electrical and Electronic Equipment.

STANDARDS

MILITARY

MS91000 —Brushes, Electrical Contact, Cartridge Type.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Requirements.** Requirements shall be in accordance with Specification MIL–D–24, and as specified herein.

3.2 **Design and construction.** Dynamotors shall be of the design, construction, and physical dimensions specified on figure 1.

3.2.1 *Circuit arrangement.* The circuit arrangement shall be as specified on figure 2.

3.2.2 *Rated voltages and currents (nominal).* The nominal rated voltages and currents shall be as follows:

Input:
Voltage .....28 direct-current volts (vdc)
Current .....3.8 amperes (amp)

Output:
Voltage .....580 vdc
Current .....0.100 amp

3.2.3 *Range of operating voltages.* The range of operating voltages shall be 22 to 32 vdc, inclusive.

3.2.4 *Duty cycle.* The duty cycle shall be continuous.

3.2.5 *Rated life at duty cycle.* The rated life at the duty cycle shall be 1,000 hours, minimum (min).

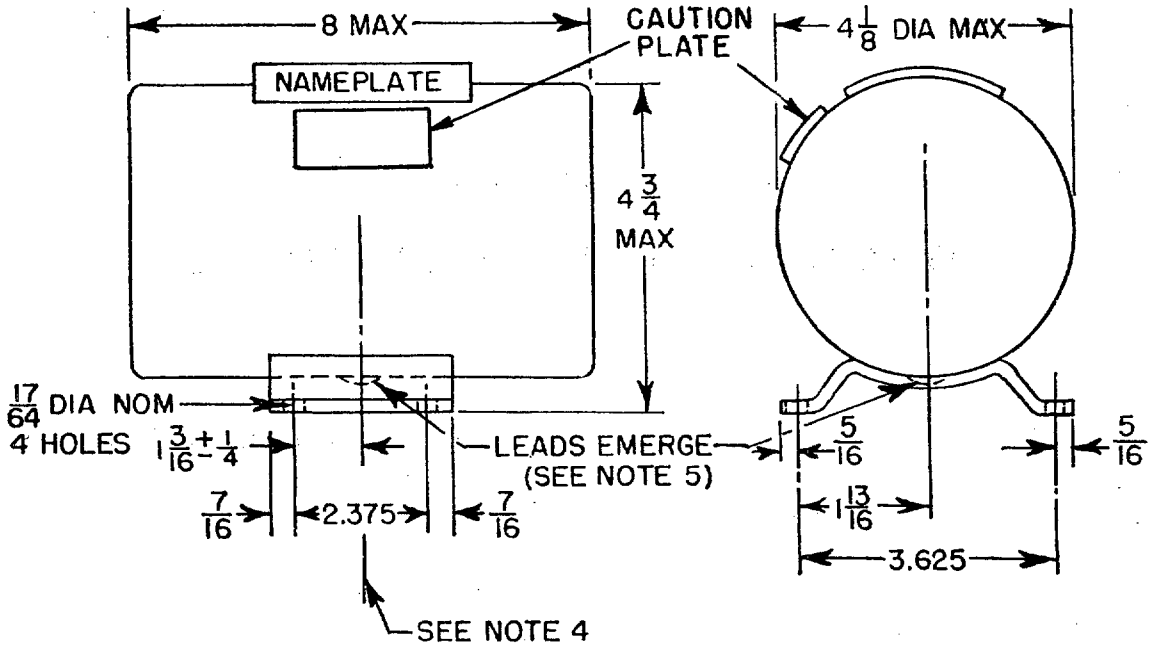
3.2.6 *Rated speed.* The rated speed shall be approximately 5,000 revolutions per minute.

3.2.7 *Ambient temperature.* The ambient temperature shall be 85° C., maximum (max).

3.2.8 *Connectors.* Connectors shall be four wire leads.

FSC 6125

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1. All dimensions in inches.
2. Overall dimensions include all permanent protrusions such as boltheads.
3. Unless otherwise specified, tolerances are ± 1/16 on fractions and ±.005 on decimals.
4. The centerline of the mounting holes shall be within ¼ inch of the center of the dynamotor overall length.
5. The dynamotor shall be provided with 4 leads of insulated, stranded copper wire, 8 ± ½ long. The ends of the leads shall be lead-alloy coated. The leads shall emerge from the dynamotor in approximately the position shown.
6. Designers of new equipment in which this dynamotor is to be used are cautioned to provide for maximum allowable dimensions.
7. The end of the dynamotor which houses the fan and the direction of rotation shall be indicated on the nameplate.
8. Capacitors should be chosen for maximum operating temperature of the dynamotor and shall not have metallic jackets.
9. Not for airborne use over 10,000 feet.
10. Caution plate shall be located so as to be readily visible and where possible as close to the output terminations as practicable.

FIGURE 1. *Type DY–134/GRC–9X dynamotors.*

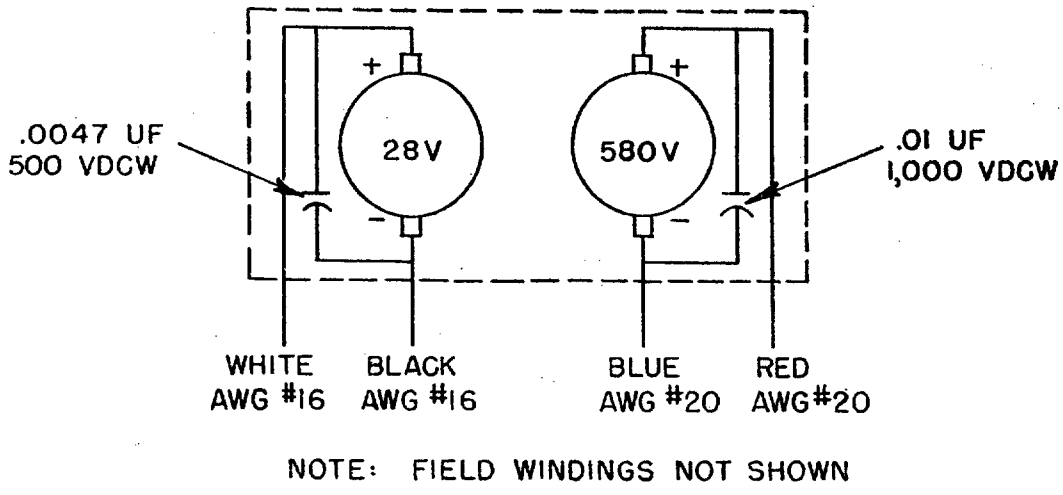


FIGURE 2. *Circuit diagram.*

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3.2.9 *Brushes.* The input and output brushes shall be as follows:

Input	MS Part No. MS91000–15PC
Output	MS Part No. MS91000–13MB

3.2.9.1 *Brush life.* The brush life shall be as follows:

Sea level	1,000 hours, min

3.2.10 *Weight.* The weight shall be 9.4 pounds, max.

3.2.11 *Enclosure.* Dynamotors shall be open protected (fan cooled).

3.3 **Radio-frequency interference.** The radiated and conducted interference shall be in accordance with Specification MIL–I–11748 (see 4.2).

3.4 **Load (see 4.3).** The load shall be as follows:

Output voltage	500 ±30 volts
Efficiency	55 percent, min
Output-voltage regulation.	15 percent, max
Ripple voltage	5 volts, max

3.5 **Temperature rise.** The temperature rise shall be as follows:

Winding	Not applicable
Frame	15° C., max

Custodians:

Army—EL
Navy—WEP
Air Force—11

Review interest:

Army—MO, MU, EL
Navy—WEP
Air Force—11, 17, 82

User interest:

Navy—MC

4. QUALITY ASSURANCE PROVISIONS

4.1 **Sampling and inspection.** Sampling and inspection shall be in accordance with Specification MIL–D–24, and as specified herein.

4.2 **Radio-frequency interference.** Dynamotors shall be tested in accordance with Specification MIL–I–11748. The test distance shall be 2 feet (see 3.3).

4.3 **Load.** Full-load measurements shall be made between the high-voltage leads with input voltage adjusted to 29.0 vdc and output current adjusted to 0.100 amp (see 3.4).

4.4 **Vibration.** Dynamotors shall be tested in accordance with test I.

4.5 **Shock.** Dynamotors shall be tested in accordance with test I.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with Specification MIL–D–24.

6. NOTES

6.1 **Notes.** The notes specified in Specification MIL–D–24 are applicable to this specification.

Preparing activity:
Army—EL

Project No. 6125–0081–9

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